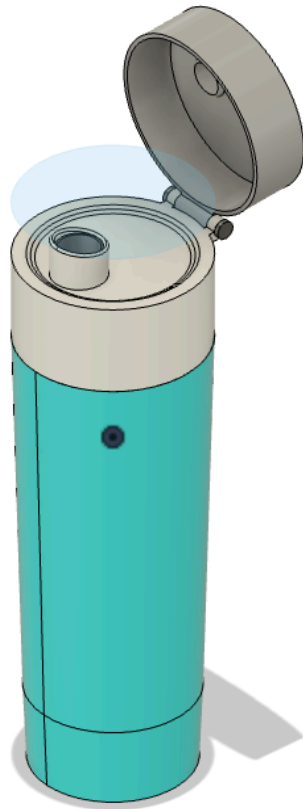




HotSip

Heated Water Bottle

ME170 Design Team
AB3-6

Team Members

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Date : 12/13/24

1.0 - PRODUCT DESCRIPTION

When considering avenues for our project, we considered the cold Illini winters that have already started. Our team all agreed that in the cold, we all enjoy a warm cup of hot chocolate, coffee, or tea during the cold winter months. During our HCD interviews, we interviewed a few college students and came to two conclusions. Primarily, all our interviewees agreed with our sentiment, they two enjoy a cup of cocoa, coffee, or tea during winter months. Secondly, they expressed annoyance with the fact that their warm beverage of choice didn't stay warm for long periods of time. With this information, we came to the question: How might we keep warm beverages warmer for longer periods of time?

With the information we gathered through interviews, we decided to create our product: HotSip. The HotSip is a heated water bottle that uniformly heats any liquid. Our product solves the issue of not having coffee, tea, etc. that stays warm throughout the day and a cold day. Our problem is experienced by students, professors, construction workers, ice sports coaches, hockey/figure skating parents, office workers, and more as they all desire to stay warm in cold climates for long periods of time and want their beverages for long durations.

We started our ideation process for HotSip by getting inspiration from popular water bottle brands and trying to keep in mind the aesthetics of our design. One of the primary reasons we wanted to keep aesthetics in mind was because one of the main motivators when buying water bottles is its aesthetic qualities. Through our ideation process, we had multiple ideas of how we would heat our bottle. Some early iterations of our design featured a screw-on battery, which proved to be unrealistic during early stages of modeling. Through conversation and further CAD design, we landed on a design that would realistically heat up the bottle that we named HotSip.

The HotSip has a polypropylene shell that gets melted together during the manufacturing process and is created using plastic molding. Inside the polypropylene shell lies the stainless steel cup which contains a column in the center of the cup which is created through progressive die stamping. The way that the cup gets heated up is by a copper coil housed inside the column at the center of the cup. The copper coil will be attached to a battery that is housed in the bottom of the shell. The placement of the copper coil makes sure that the heat is uniformly distributed in the liquid. The lid design is made up of polypropylene as well with a stainless steel lining inside the bottom of the lid ensuring the liquid only touches the stainless steel and not polypropylene. The lid uses a hinge mechanism to keep the liquid warm by not allowing the heat to escape when closed as well as keeps the liquid where it should be: in the cup.

Since this product is made for beverage consumption, we made sure to take into consideration FDA regulations for food-safe materials. That is why we chose stainless steel as the cup where the drink will be housed as it is a food-safe material. Further, stainless steel will keep the heat inside the drink. Because stainless steel retains heat and hot stainless steel would burn skin on contact, we decided to create the polypropylene shell which would not burn skin on contact but keep the heat inside the liquid not on the hands.

As stated above, our product will be useful for all people of various ages and backgrounds. Some examples of potential target users of our product could be college students in a cold environment, people who enjoy a warm beverage, construction workers who work in the cold, winter sports coaches, and more. Besides that, our product is able to be used by a variety of people and is not restricted to a certain age group, background, or more. Our product can be universally used even though the design was based on the unmet needs faced by college students. However, our product is not limited to college students as everyone can face the same issue.

Consumers should buy our product, the HotSip, because it solves the issue of not having a warm beverage that remains warm even for long periods of time. Also our product, in comparison to the competition, will be able to be sold for a lower price. Our product's competition is sold for around eighty dollars and with our choice of material, we would be able to sell our product for less while still making a profit, which initially would cover the cost of machining.

To finish off, our product HotSip will allow many people of various backgrounds to keep their warm beverage warm for long periods of time. Further, our product kept in mind design aspects of various popular water bottle brands like Owala, Stanley, and Hydroflask. In addition, we used FDA compliant materials to make sure our product is safe for human consumption. So keep your beverage warm with HotSip, even in below freezing temperatures.

2.0 - CONCEPT SKETCHES

2.1 - Initial Concept Sketches

Figure 1:

Initial Sketch #1

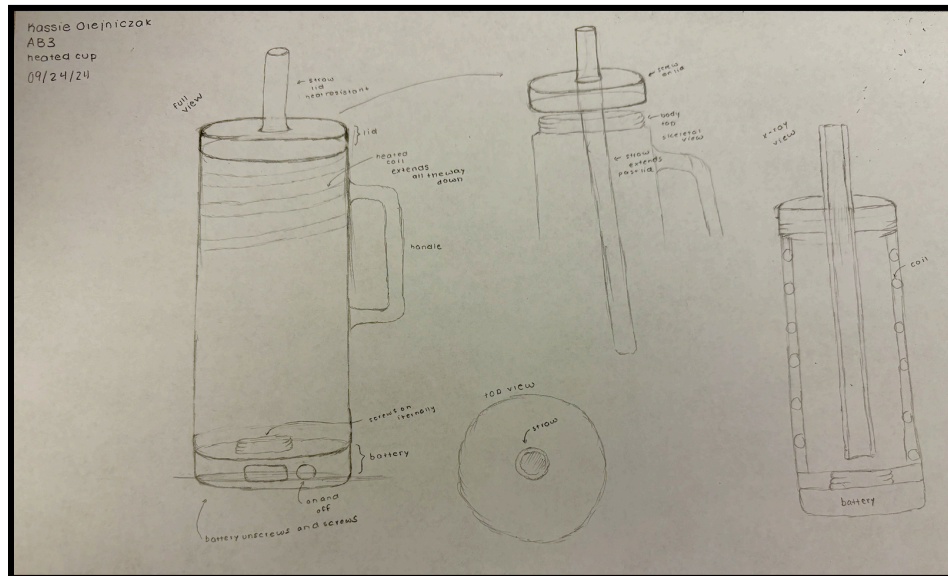


Figure 2:

Initial Sketch #2

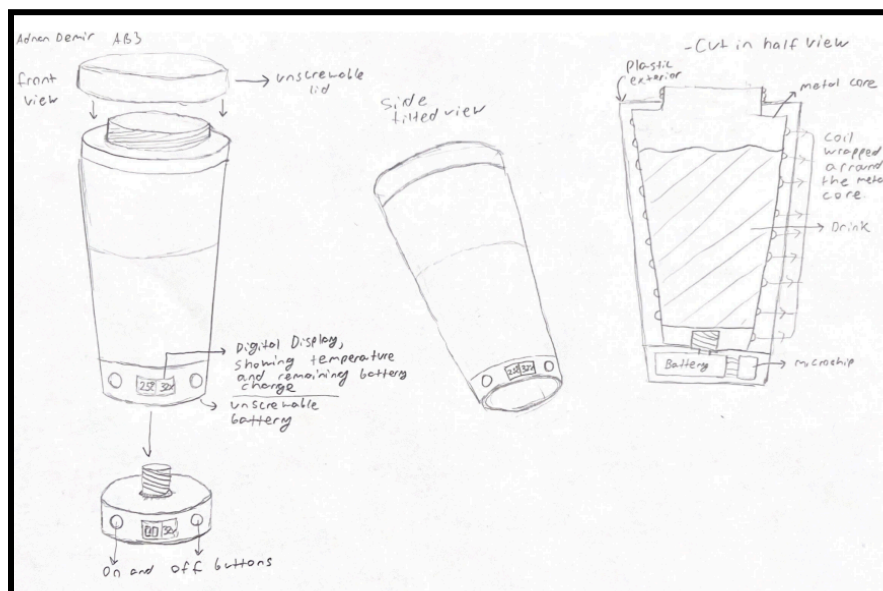


Figure 3:
Initial Sketch #3

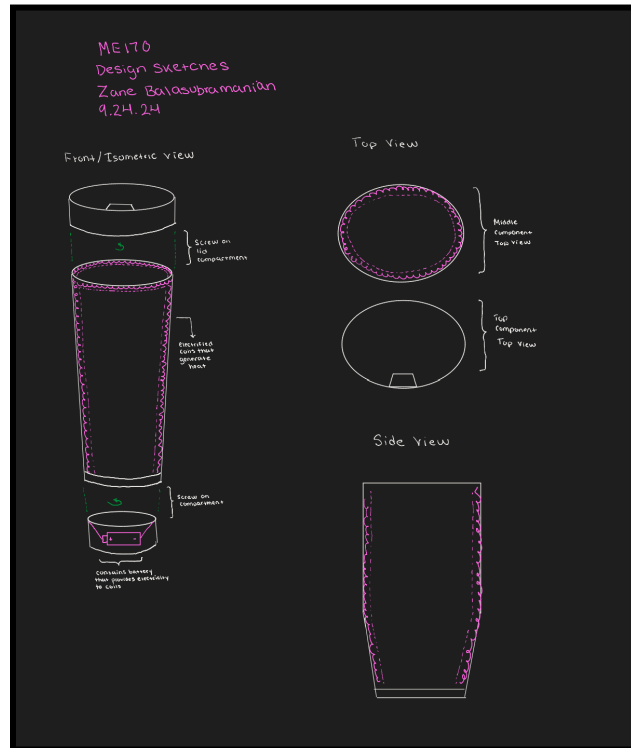
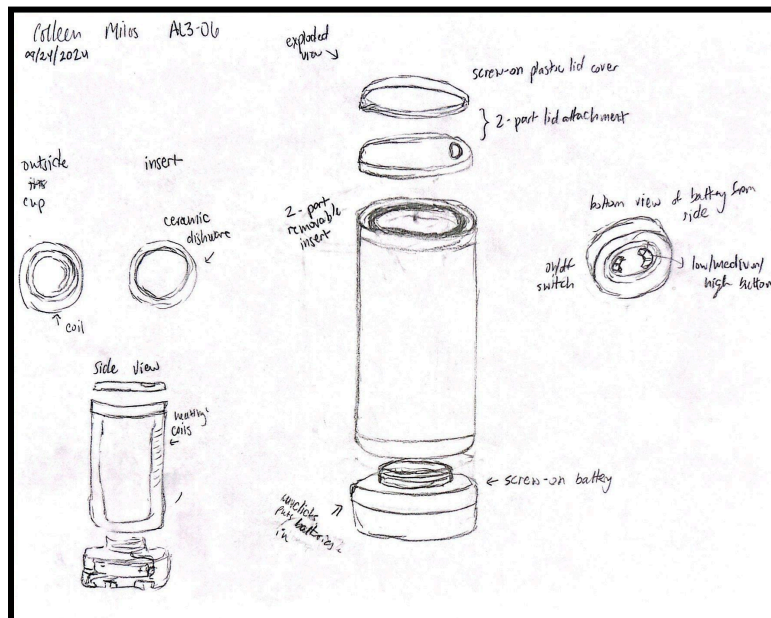


Figure 4:
Initial Sketch #4



2.2 - Initial Concept Sketch Narrative and Transition to Final Product

Our initial ideation and Pugh matrix designs (see figures above) are far different from our final product and rendition of the HotSip. In this initial phase, we took fundamental inspiration from various popular thermos and bottle designs and proceeded to make alterations so that our sketches fit the niche problem space for our particular product. While each of our several designs reflected a possible method of approaching this problem space, they differ greatly in how they operate.

Our first possible design reflected a structure inspired by Stanley Cup products; it featured a straw and handle mechanism (much like the product we took inspiration from), but further contained a copper coil and battery system that could heat the liquid when an “on” button was pressed near the bottom of the bottle. Our second design rather broke the bottle into three screwable components (i.e., the lid, body, and battery section); the lid could be screwed on and off of the main body by the consumer when they wished to drink the liquid, and the bottom battery component could be detached so the battery itself could be replaced. This design also contained a digital interface near the bottom of the bottle that could display to the user various informative properties (i.e., the remaining battery life, temperature, etc.) The third design reflected a simplification of the two prior; it also contained three screwable components, but additionally included a slit on the lid that the user could directly drink from, as opposed to screwing the lid off every time. This design also contained a removable battery component, but it did not include a digital interface; it also did not utilize a handle, rather using insulative materials so the thermos was safe to touch. Our fourth design reflected a difference in structure from the designs prior, but retained the heating properties and methodology; as seen in the figure above, this design instead utilized various locking / clicking mechanisms in addition to screwable components to make the design more easily accessible and user friendly.

As we proceeded to narrow down our initial designs and move towards selecting a final method of approach, we had to take into consideration various areas of constraints we had not initially considered; a major one of these areas was safety and FDA compliance. We realized how unrealistic our screw-on battery idea was in terms of operating and connecting it to the copper wire; not only would this be difficult to design, but it would raise many safety concerns and be potentially dangerous to our consumer (having the ability to cause burns). We further had to adjust our initial designs and take into consideration the proper insulation and space necessary for the product to not only realistically heat the liquid, but especially do so in a safe manner; many aspects of our initial designs would not realistically accomplish this. Another important area we had to adjust our designs for was user-interface; many of our initial lid sketches and methodology for how the user holds / interacts with the product raised issues of convenience and operability- would it really be realistic, for instance, to have our user unscrew the lid every time they wished to drink? Or, if not using the screw-mechanism, would they really prefer to drink a heated beverage from an extremely small slit? These are all questions and areas we began to consider as we moved toward revising our initial designs and selecting a final approach to move forward with.

3.0 - CONCEPT SELECTION PROCESS

3.1 - Pugh Matrix

	Weight	Design 1 (Kassie)	Design 2 (Adnan)	Design 3 (Zane)	Design 4 (Colleen)
Safety	9	0	0	0	0
Aesthetics and Finish	8	1	1	1	1
Material	6	0	-1	1	1
Performance	5	0	1	0	1
Service Life	3	-1	1	1	-1
Manufacturing Facility	7	1	0	1	-1
Standards, Specifications, and Legal Aspects	8	1	1	0	1
Customer	6	0	0	1	1
Disposal	4	-1	1	0	-1
Weight	7	1	1	0	1

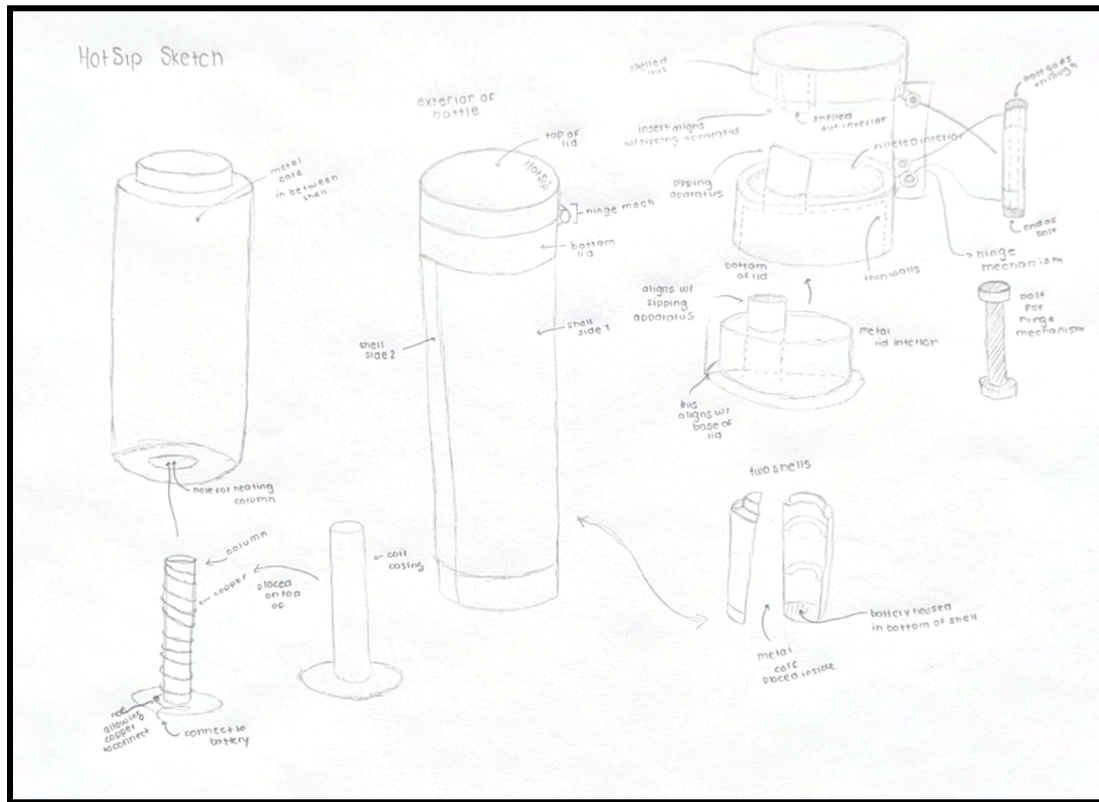
Total q of +1's	4	6	5	6
Total q of 0's	4	3	5	1
Total q of -1's	2	1	0	3
Overall Weighted Score	23	29	29	26

3.2 - Final Design Selection, Iteration, and Concept Sketch

3.2.1 - Final Design Concept Sketch

Figure 5:

Final Design Sketch including iterations



3.2.1 - Final Design Description and Iteration

It is important to note that while we utilized the above matrix to select a method of approach from our initial designs, our chosen design underwent frequent revision and changes to bring us to the final concept iteration described below (and pictured in the image above). We continued the process explained in section 2.2 to narrow down and include additional criteria; this criteria is outlined in depth in section 4.0.

As seen above, our product changed a lot from initial design sketches to our final product. While we began with the fundamental ideas presented in initial sketches 2 and 3 (those that scored highest in the pugh matrix), we made many further changes as we continued with the design process. The main reasons associated with such a drastic change of design came from when we discussed the actual functionality of our product. Our initial concept designs were focused primarily on aesthetics and following water bottle trends. Further, our initial concept of a screw on battery was not conducive to the actual functionality and practicality of our product. Because of that we changed almost the entirety of the design to make our product functional. While our knowledge of how a battery can be operated in our context, we knew that the screw on the battery would not connect to the copper coil effectively and correctly to heat our product.

Because of this we created the copper coil in the center of our bottle and created a shell where the battery can be housed without affecting the functionality of our product and without affecting the liquid inside the cup. Our final product was able to be both functional and practical.

4.0 - PRODUCT DESIGN SPECIFICATIONS

Our team has elected to design a water bottle that is capable of producing its own heat to keep its contents warm. The bottle would be travel sized and portable through everyday use.

1. Performance

- Speed - The product itself does not need to be fast or slow as our product is stationary. However, one of the key features of our product is its ability to heat up liquid. The product needs to heat up and retain the heat created. The product does not need to heat up terribly fast, however, it can also not take forever to heat up. Desired “heat up” time should be around 3-4 minutes.
- The cup would heat continuously, until the off button is pressed. Once the cup reaches peak temperature, the amount of heat would plateau.

2. Environment

- We decided that the battery should supply 130 degree fahrenheit max heat to the liquid. In contrast the product itself should be able to exist in any temperature range, from cold temperatures to hot temperatures.
- Our product needs to be able to handle a degree of abuse as people of any age can handle this product and typical water bottles do sustain a degree of damage and dents throughout their use.
- Furthermore, this product needs to be durable enough to withstand the same environments that a water bottle would be used in (examples include: campgrounds, office, work out facilities, outdoor and indoor facilities). However, in the context of our product, we would argue that our product should be able to withstand cold temperatures, as that would be where it is ideally used.

3. Service Life

- With this type of product, I think that there would be a short service life due to the fact that, if the product is available at retailers and other high demand outlets like Amazon, a short service would be better to turn our products in an efficient time
- The average service life/life span of a typical thermos is 5-10 years

4. Maintenance

- The only maintenance on the product that would be needed reflects the same maintenance needed for a water bottle or mug. This type of maintenance is just basic cleaning after use. The customer should expect similar maintenance to a stereotypical hand wash water bottle would.
- The maintenance for the product is easy as it just needs to be run under water and soap which is readily available in most homes

5. Target Costs

- Heated mugs range from \$40-\$100 depending on the quality and brand with majority of the products being closer to the hundred dollar range, sitting in the upper 70 to upper 80 dollar range.
- Ideally, the product could be priced in below the upper 70 to upper 80 dollar range to account for the fact it acts more closely like a water bottle than a heatable mug and our product being more portable than the mug.
- With mass production a sufficient profit should be made as well as accounting for any labor and supply costs.

6. Competition

- Our competition is the conventional thermos which does not keep your beverage warm for long periods of

time (the main grievance we aim to solve) and then heated mugs which aren't as portable as water bottles and thermoses are. Those heated mugs and thermoses stay in the upper 70 and 80 dollar range,

- To differentiate ourselves from competition, we will need to have a different design and look. Also our mode for heating up the bottle is different and guarantees uniform heat distribution based on our column heating module.

7. Shipping

- We believe that the product could be bulk shipped to retailers like Target, Walmart, and other retailers. Further, the product could be shipped through Amazon.
- Because of the size, the shipping costs would not be incredibly high due to its size and potential weight.
- Average weight of thermoses is about 2.5 pounds, however, with our design's inclusion of a battery, we can predict that our product would weigh more than the standard 2.5 pounds that normal thermoses are.

8. Product Volume

- We are aiming to produce this product on a larger scale in higher volume quantities once we achieve a steady stream of consumers / market, and further sell this product to large chain stores such as Target, Walgreens, Walmart, etc. and sell this product to large chain stores in bulk. We aim to produce 100,000-500,000 parts annually.
- In this sense, we want to design with DFM (design for manufacturability) in mind, choosing manufacturing processes that have a low cost per part and as little labor costs as possible.

9. Packing

- The product will not require any additional packing components beyond what is typical, as it has no extenuating conditions that would require a certain type of packaging.
- The product will likely be produced/shipped to the consumer fully assembled, with a battery attached in the bottom of the polypropylene shell. Our product will be produced at a facility where all the components will be put together and then shipped out to our distributors or to consumer homes.
- In general, we will want to have some sort of foam, packing peanuts or paper within the packaging to avoid the bottle being dented or harmed in any shipping process. These are very minimal and low cost materials however; i.e., recycled crinkle paper / additional cardboard should cost somewhere between \$0.20 to \$0.40 per shipped product. The actual shipping and packaging relative to the box will depend on where we are shipping.

10. Manufacturing Facility

- Our product can be manufactured in existing plants as there are a large variety of facilities with the pre-existing machinery necessary to produce it. (i.e., similar cups have been designed, as have thermos' with closable lids).
- There are a variety of adjustments that will need to be made beyond general sizing/dimensions, including space for the heated column and various parts made for the lid. This might require some new machinery for certain parts or aspects of the product.

11. Size

- The product must fit a variety of size constraints based on general surveyable preferences and also designability constraints:
 - The product's main liquid compartment must be able to store enough liquid and design should be able to be modified to be scaled up or down as many popular water bottle brands do. Our product must be able to hold enough liquid to be considered a drink (i.e., 16 oz-20 oz).
 - The liquid must be small enough in volume to be reasonably heated via our coil column method, depending on the electric capacity of the coil. (i.e., 100 watts, 500 watts).

- The product must be small enough size to be considered portable and desired by consumers.

12. Weight

- The product must fit a variety of weight constraints primarily based on consumer standards/preferences, but also relative to shipping costs.
 - When considering portability and user preferences, the cup should not exceed over three pounds in weight; there are a variety of components of the product and these should be totalled together when considering weight. There is also consideration of the liquid itself, although it is not part of the three pound ideal based on the components of our bottle.
 - Shipping and user preferences hold the same ideals in that neither want the product to be too heavy; if the product stays under 3 pounds, while considering the additional weight of the package and filler, shipping should not exceed an unreasonable amount in costs.

13. Aesthetics and Finish

- The finish should be extremely smooth with little to no texture on the surface for aesthetic purposes. The polypropylene that the shell and lid should be smoothed out and the stainless steel on the inside should be smooth. The polypropylene should not be misshapen. There is little room for error in finish, as aesthetics are important to the consumer relative to our product.
- The color of the product can be variable but should be uniform throughout the product.
- The product should adhere to all designed dimensions and constraints.

14. Materials

- A variety of materials should be used when producing this product to meet its design requirements. The polypropylene shell and lid will make sure that the liquid is insulated and does not cause burns. Further, for the actual part of the product that is in contact with the liquid that will be consumed by consumers, we decided to go with stainless steel as it is FDA compliant and what other water bottles are made. Further, the coil is made out of copper to conduct heat that will be heated and heat up the stainless steel cup cover.

15. Product Lifespan

- Given the lifespan of comparable products such as slow cookers and large insulated tumblers, our product should withstand at least 10 years. Slow cookers date back to 1971 and Thermos brand stainless steel mugs date back more than 100 years.
- However, given the current market's capacity to sell water bottles and tumblers as 'microtrends', each water bottle on the market only gets incredibly popular around a year and a half.

16. Standards, Specifications, and Legal Aspects

- Within the FDA's Code of Federal Regulations Title 21, hazardous substances and indirect food additives are restricted and heavily monitored within stainless steel bottle production. Hot-melt strippable food coating and different types of vinylidene nylon film coatings are restricted.
- Substances such as lead, cadmium, mercury, and phthalates have been restricted from consumer products according to California Proposition 65, as imposed by the California Office of Environmental Health Hazard Assessment.
- A Country of Origin label must be permanently affixed within the stainless steel bottle.
- As we are sticking with a plastic screw cap design, the commission regulation no. 10/2011 as of January 14, 2011 has provided a set guideline of boundaries restricting plastic materials and articles intended to breach contact with food. PFA's, Nitrosamines, and Polyvinyl chloride are restricted and cannot migrate into food at a harmful level.

17. Ergonomics

- The cup will be able to withstand 6 drops without breaking.
- An average person will be able to operate the bottle. Opening and closing the lid and unscrewing the lid and turning on the battery.

18. Customer

- After interviewing many people regarding the concerns with water bottles and their ideal water bottle/thermos, we've decided that the bottle can't carry more than 32 oz of liquid inside and no less than a typical cup of coffee.
- For future iterations and considerations we should create an additional component that resembles a handle for easy transport and additional functionality.
- The design cannot be glaringly loud or abhorrently industrial; it has to appear to other water bottles that are really popular brands on the market.
- The outside touch of the bottle can't be incredibly warm, there should be little risk of injury, specifically burns, when operating our product. Hence the introduction of the polypropylene shell that is part of our design.

19. Quality and Reliability

- The material that should be used on the interior components of the bottle should be 18/8 or 304 stainless steel. This is consistent with other types of stainless steel products.
- The shell should be made out of polypropylene because it will keep the heat on the inside and liquid, reducing risk of severe burns on human skin.
- The manufacturer must meet ISO 9001 standards for quality management as well as ISO 14001 for environmental management.
- The bottle should be LFGB certified, which ensures its safe for food contact.

20. Shelf Life

- The shelf-life of this bottle should be around 5-10 years, which would be consistent with other kinds of stainless steel heated mugs on the market.
- The design of our bottle with the implementation of Lithium-ion batteries still render it reliable, but the aspect of charging and/or replacing batteries does make it less attractive from a retailers' standpoint.
- The operator will have to consistently replace the batteries with standard, single-use batteries. Given functions that regulate heat and don't consistently run within the bottle, these batteries will withstand a substantial amount of time.
- As our design progresses and demand rises, options for rechargeable batteries will be introduced.

21. Processes

- The processes our design will follow will be the mainstream, utilizing a polypropylene shell and a stainless steel bottle inside the shell. The battery will be at the base of the polypropylene shell which will be connected to the copper coil which supplies heat uniformly in a stainless steel column.
- Our bottle components will be stainless steel made with progressive die stamping and the polypropylene shells will be created through injection molding. The components of the interior of the shell will be assembled and the two shells will encase the interior and then be melted together.
- We also should consider going through with electrolysis polishing to ensure smoothness of the shell and the stainless steel interior.

22. Timescales

- After establishing our brand name and market analysis, we will financially plan and secure funding through personal investors and business loans.
- A government-issued licence and documentation such as DBAs, EINs, ITFs, and articles of incorporation

will be required.

- All in all, this process should take around 3 years to secure all the necessary funding, open up means for production and contact retailers for sale opportunities.

23. Testing

- The product will be tested in at least 5 prototype designs before mass production and release and its electrical components will be tested to avoid overheating.
- Rigorous testing will be conducted to ensure that the heat preserving/isolating material works as intended and does not leak much heat and harm the user.

24. Safety

- Heat insulating materials, the polypropylene shell in our case, should be 2 mm thick and placed a distance from the main heated component of our design (that being the stainless steel cup/core). This is done to avoid injury and other harm. It is important to know that the Standard Guide for Heated System Surface Conditions that produce contact burn injuries recommends that the surface temperature of a customer product remain below 140 degrees Fahrenheit as humans can touch this temperature for up to 5 seconds without sustaining irreversible burn damage. However, the cup would need to go further below this limit anyway as it needs to be comfortable to touch for longer durations.

25. Company Constraints

- The company must allocate a minimum of 7-8% of the total budget annually to research and development efforts aimed to improve battery life, battery handling, aesthetic improvements, and potential new developments in heating technology.
- The company must ensure that the product receives correct certifications such as the one given by Underwriters Laboratories that ensures electronic products are up to standards. In addition, we must make sure that we have the correct documentation required to operate a business and keep our product FDA compliant since our product is made for liquid consumption.

26. Market Constraints

- The product must appeal to a variety of individuals, including but not limited to: people who travel frequently, work in an office setting, or engage in outdoor activities. These are some of the groups of people we believe could benefit from a product like ours.
- The product must be available in at least 100 retail locations, initially in the United States. This would assure that it has a wide range and data could be collected as to where it sells the best. This would lead to the prioritization of such locations, increasing the effectiveness of the marketing and sales.

27. Patents, Literature, and Product Data

- Filing a patent can have varying costs but for a large company they could allocate a budget of \$15,000-\$30,000 for patent filing and legal fees to secure intellectual property rights.
- A comprehensive technical literature study must be conducted to identify the effectiveness of our design in comparison to competitors and ensure that our cup can achieve success on the wider market.

28. Political and Social Implications

- The company producing this cup must navigate and comply with varying international trade regulations and rules about import/export policies. In addition the materials used may be legal in certain countries while being illegal in others.
- An active effort must be made to promote our commitment to sustainability and consumer safety to ensure that the social implications of our product remains positive and that we can ensure our place in the market. We could achieve positive social implications by using at least 70% recycled or recyclable materials in the

construction of this product. This would reduce our carbon footprint and make our product more popular.

29. Disposal

- In most countries it is not allowed to simply dispose of a battery thus it would be difficult for a customer to dispose of our product once they no longer need it. This is why a return program could be initiated to collect used products once the customer no longer needs it, this could be accompanied by a \$5 discount for the next purchase made by the customer or another similar incentive program. A goal could be set for yearly recycling (such as 1000 units recycled per year).
- This product must be designed in a way that the metal and plastic used to make it could potentially be recycled to ensure a lessening of environmental damage and carbon footprint. This would be beneficial to the environment.

5.0 - CAD MODELS

5.1 - Assembly View CAD Model

Figure 6:

CAD model with lid closed

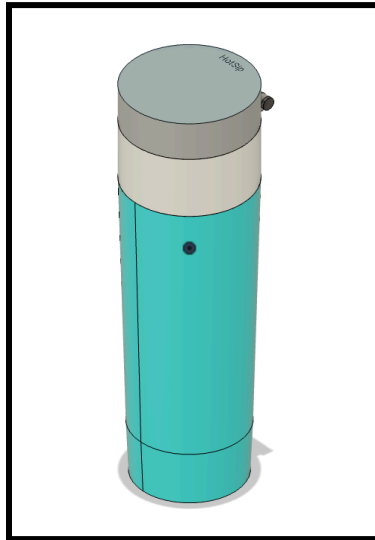
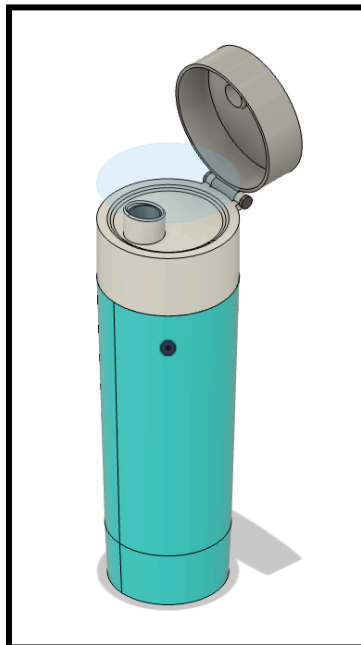


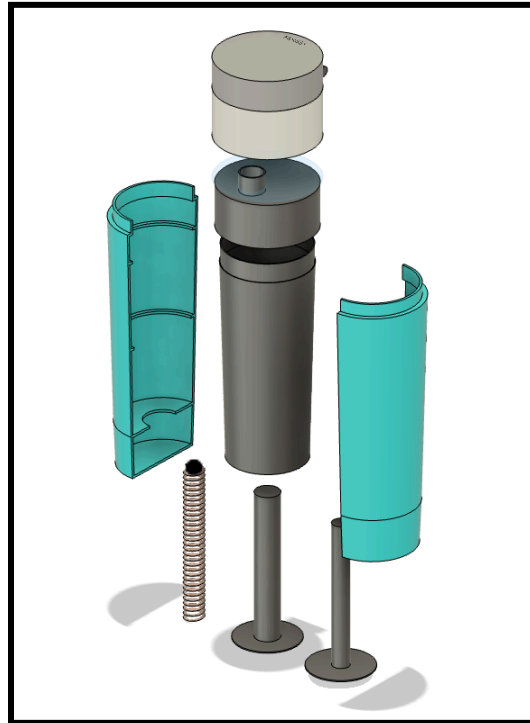
Figure 7:

CAD model with lid open



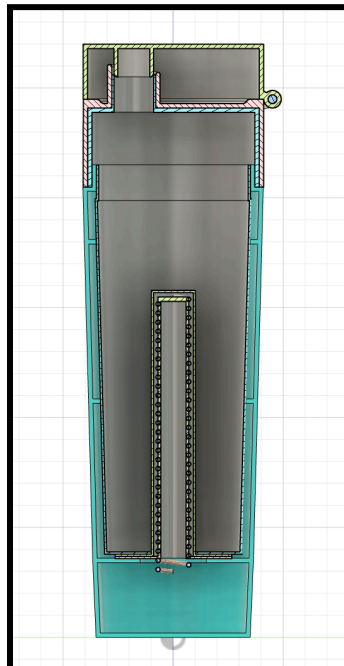
5.2 - Exploded View CAD Model

Figure 8:
Fully exploded CAD Assembly Model



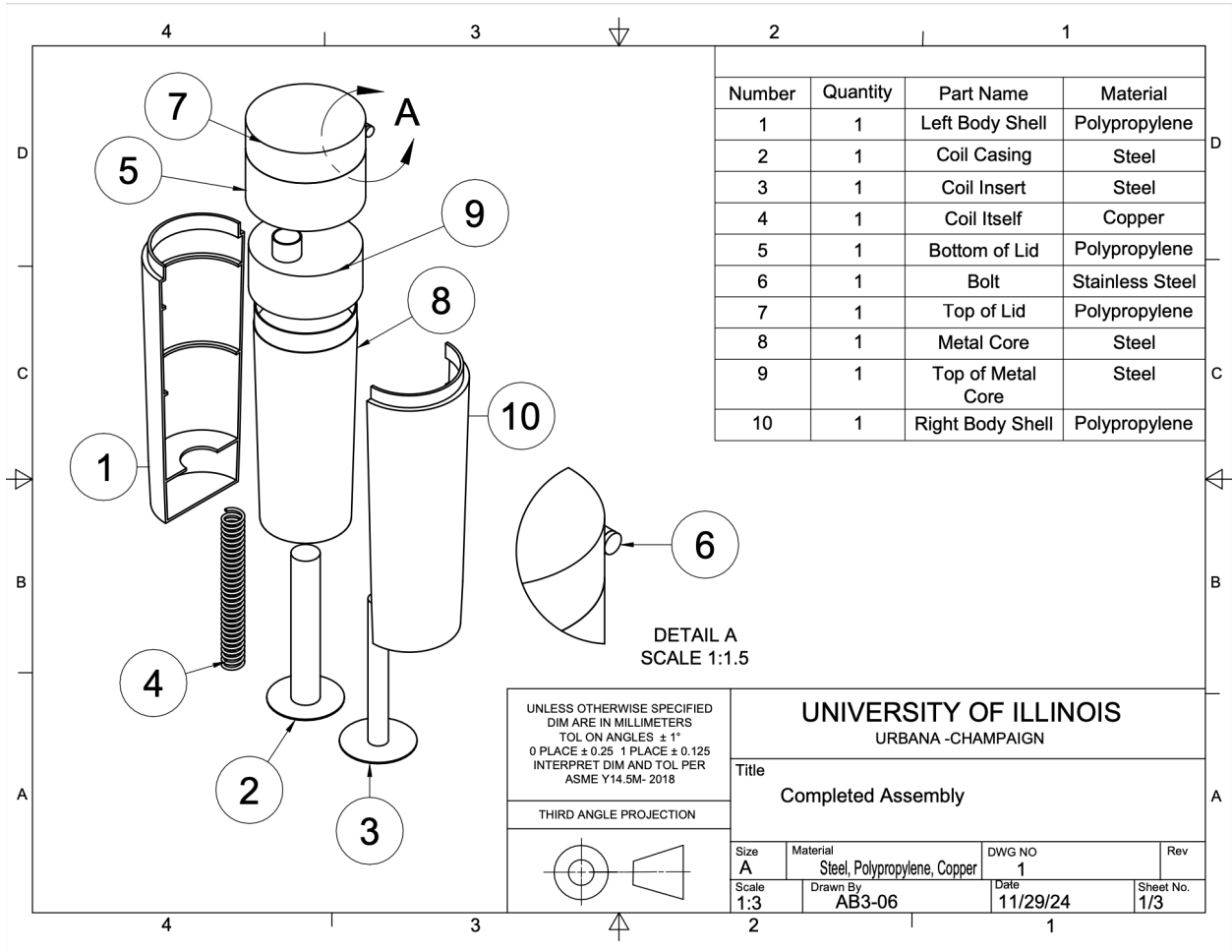
5.3 - Cross Section View CAD Model

Figure 9:
Cross-Sectional Assembly View



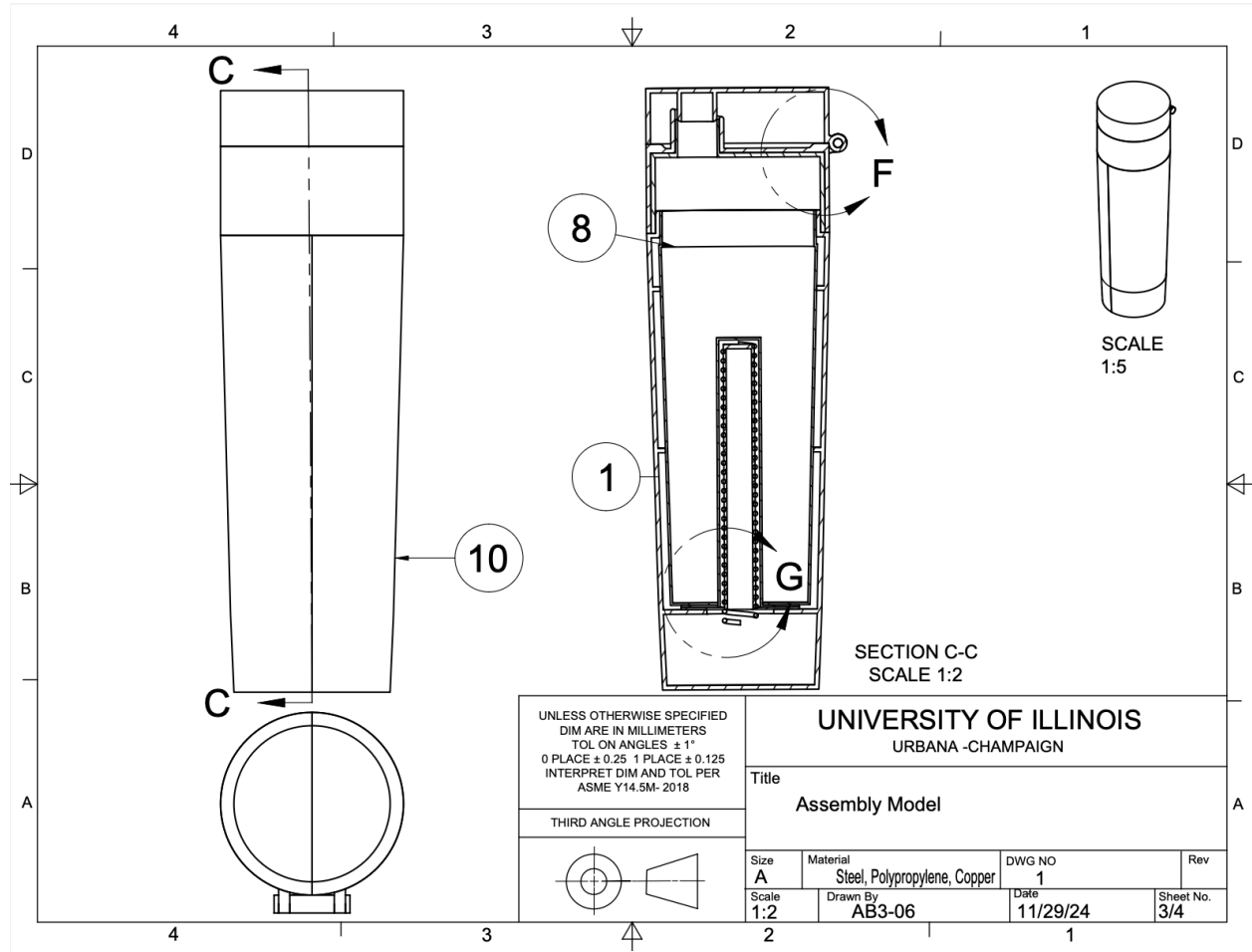
6.0 - EXPLODED ASSEMBLY DRAWING AND BILL OF MATERIALS

Figure 10:
Exploded Assembly Drawing with BOM chart



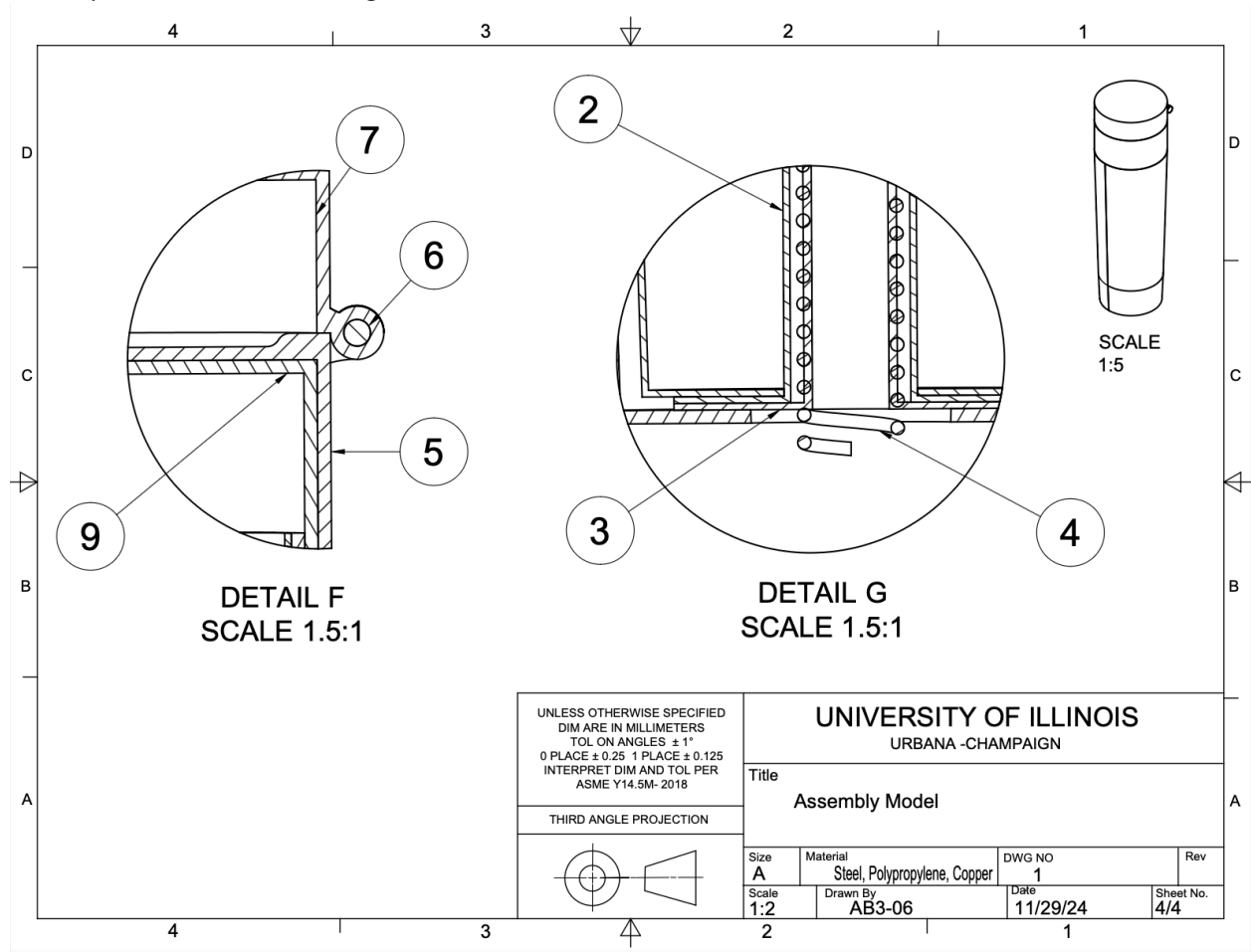
7.0 - ASSEMBLY DRAWING WITH CROSS-SECTIONS

Figure 11:
Assembly Cross Sectional Drawing - 1



This first cross section essentially shows the entire assembly. Since every part is circular and assembled on top of each other it can be easily observed with one sectional analysis. In this section it is visible how the body shell keeps the metal core, coil casing, and coil insert constrained in place by simply having them touch its inner shelves. It can also be seen how the lid attachment part keeps the lid connected to the assembly and improves the structural integrity.

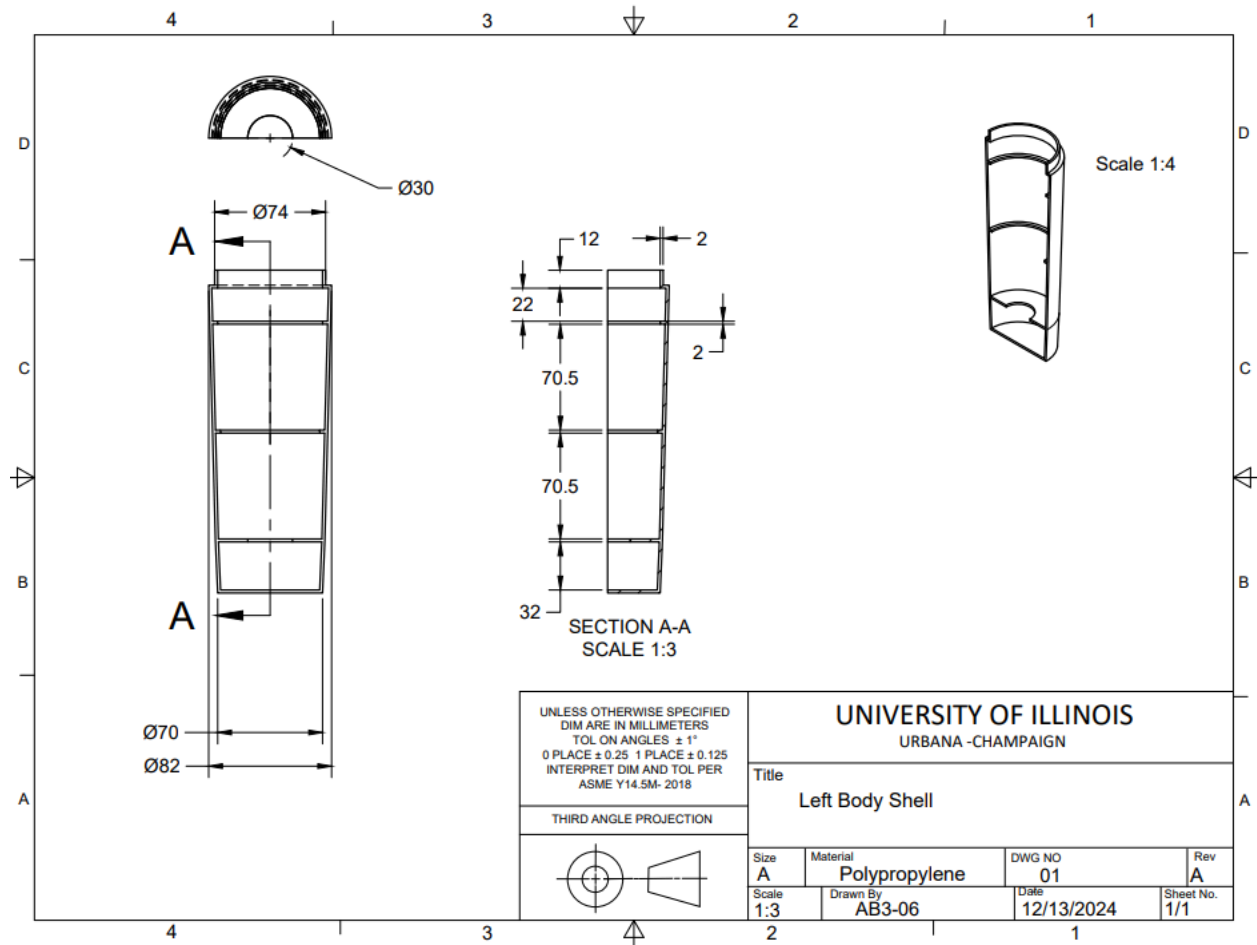
Figure 12:
Assembly Cross Sectional Drawing - 2



The second cross sectional drawing showcases the more detail oriented areas of the assembly. These being the hinge and the coil casing-insert area. The hinge mechanism and its geometry as well as the coil's movement through the assembly can be seen in detail. Specifically in Detail G, you can see how the coil column, coil, and coil casing are all stacked neatly and firmly together.

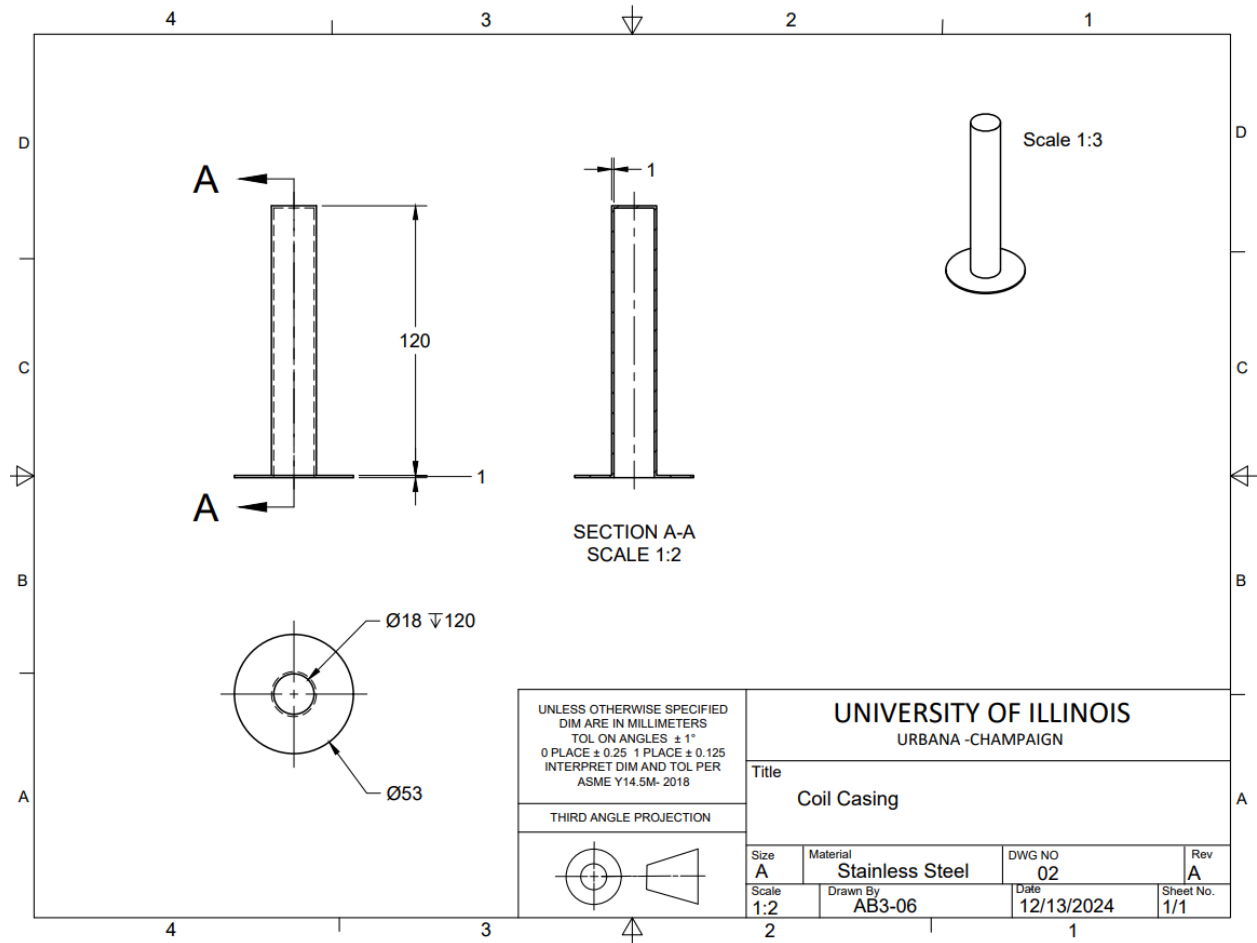
8.0 - DETAILED ENGINEERING PART DRAWINGS

Figure 1:
Left Body Shell Drawing



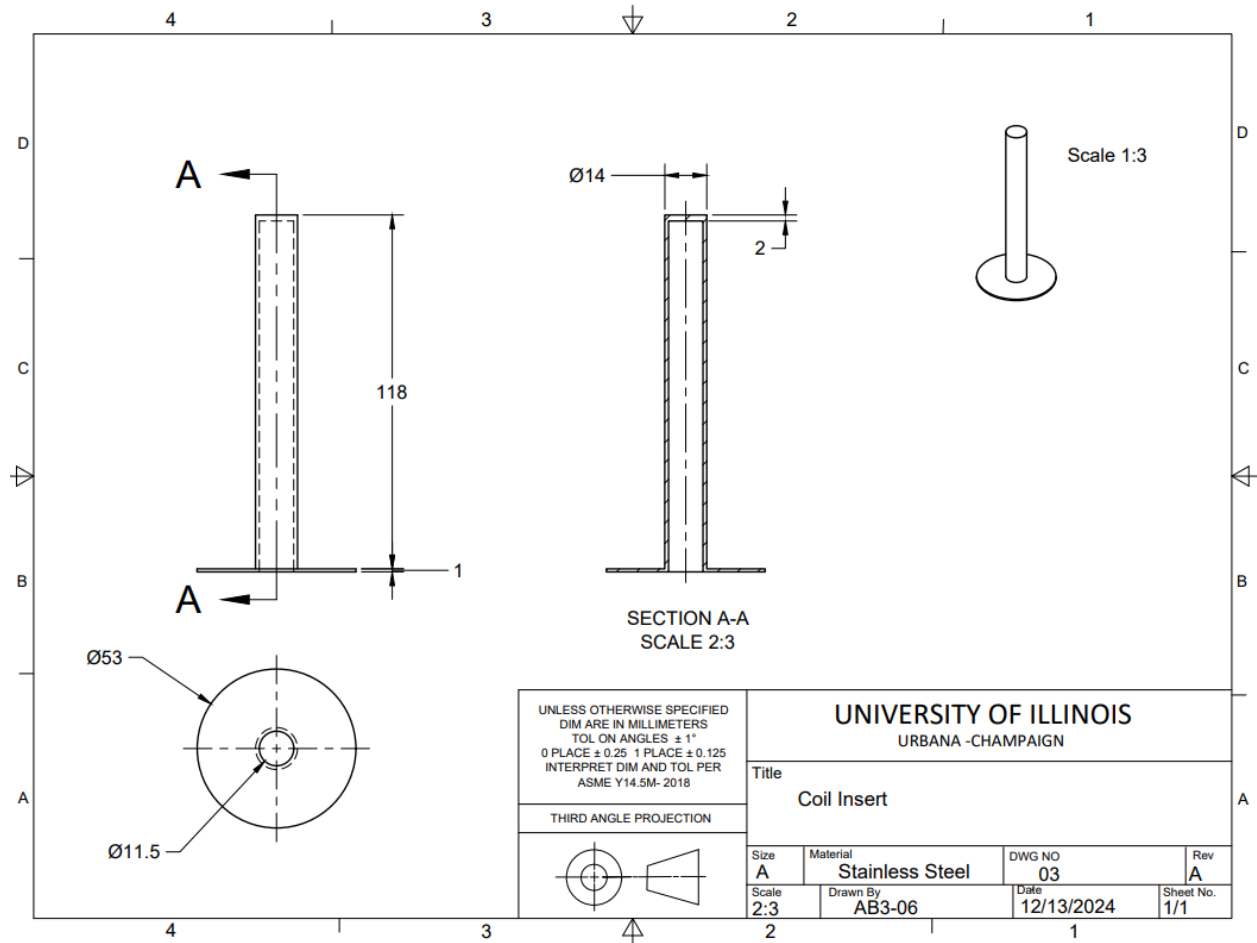
The left body shell is one of the most important parts of the project. Its central position along with the strategically placed shelves within allow it to hold the internal parts of the cup together. Once the left body shell is joined with the right it causes all internal components to be fitted into place rigidly. The top of the left body shell is designed to fit within the lid and allows for easy assembly and fitting. The left body shell also features a compartment where batteries and other electronic components could be placed. The shelf above the compartment has a large hole in it that allows the coil insert to connect to the bottom, letting the copper wire reach the battery.

Figure 2:
Coil Casing Drawing



The coil casing houses the coil insert and the coil that heats the liquid. It is designed in a way to ensure that the heat from the coil can easily pass into the drink without having the coil itself touch the liquid. The Shape chosen allows for easy manufacturability and assembly. The flat bottom circle ensures that it can fit between the left and right body shells and be constrained in place without needing a specific assembly element like screws or other.

Figure 3:
Coil Insert Drawing



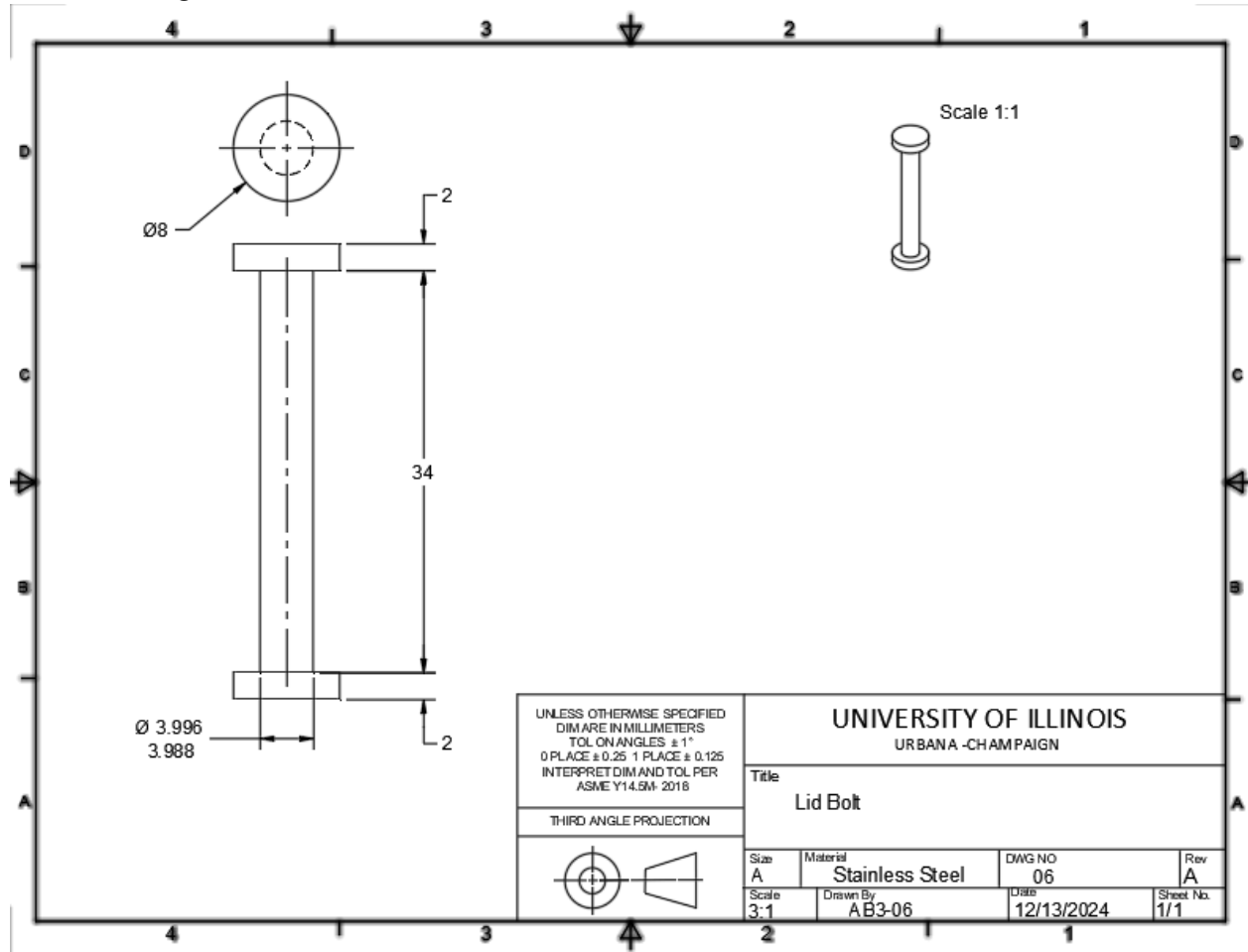
The coil insert is the part that the coil wraps around. After the coil is wrapped around the insert it is put into the lower end of the coil casing. The circular feature at the bottom has a small slit in it that lets the coil connect to the battery housed below. This part can also be simply attached to the parts above it and constrained in place by the left and right body shells once the full assembly is created.

The drawing shows a lid bottom assembly. The top view (top left) shows a circular part with a central hole of diameter Ø20 and an outer diameter of Ø24. There are two small rectangular features on the left side, each with a width of 8. The side view (top right) shows a cylindrical part with a diameter of Ø2 and a height of 4. The section view (bottom left) shows a cross-section of the lid bottom with a diameter of Ø2 and a height of 4. The section view also shows a larger diameter of Ø52 and a total height of 36. The section view is labeled 'SECTION A-A' and 'SCALE 1:2'. The title block (bottom right) contains the following information:

UNLESS OTHERWISE SPECIFIED DIM ARE IN MILLIMETERS TOL ON ANGLES $\pm 1^\circ$ 0 PLACE ± 0.25 1 PLACE ± 0.125 INTERPRET DIM AND TOL PER ASME Y14.5M-2018				UNIVERSITY OF ILLINOIS URBANA -CHAMPAIGN	
Title Lid Bottom					
Size A	Material Stainless Steel	DWG NO 05	Rev A		
Scale 1:2	Drawn By AB3-06	Date 12/13/2024	Sheet No. 1/1		

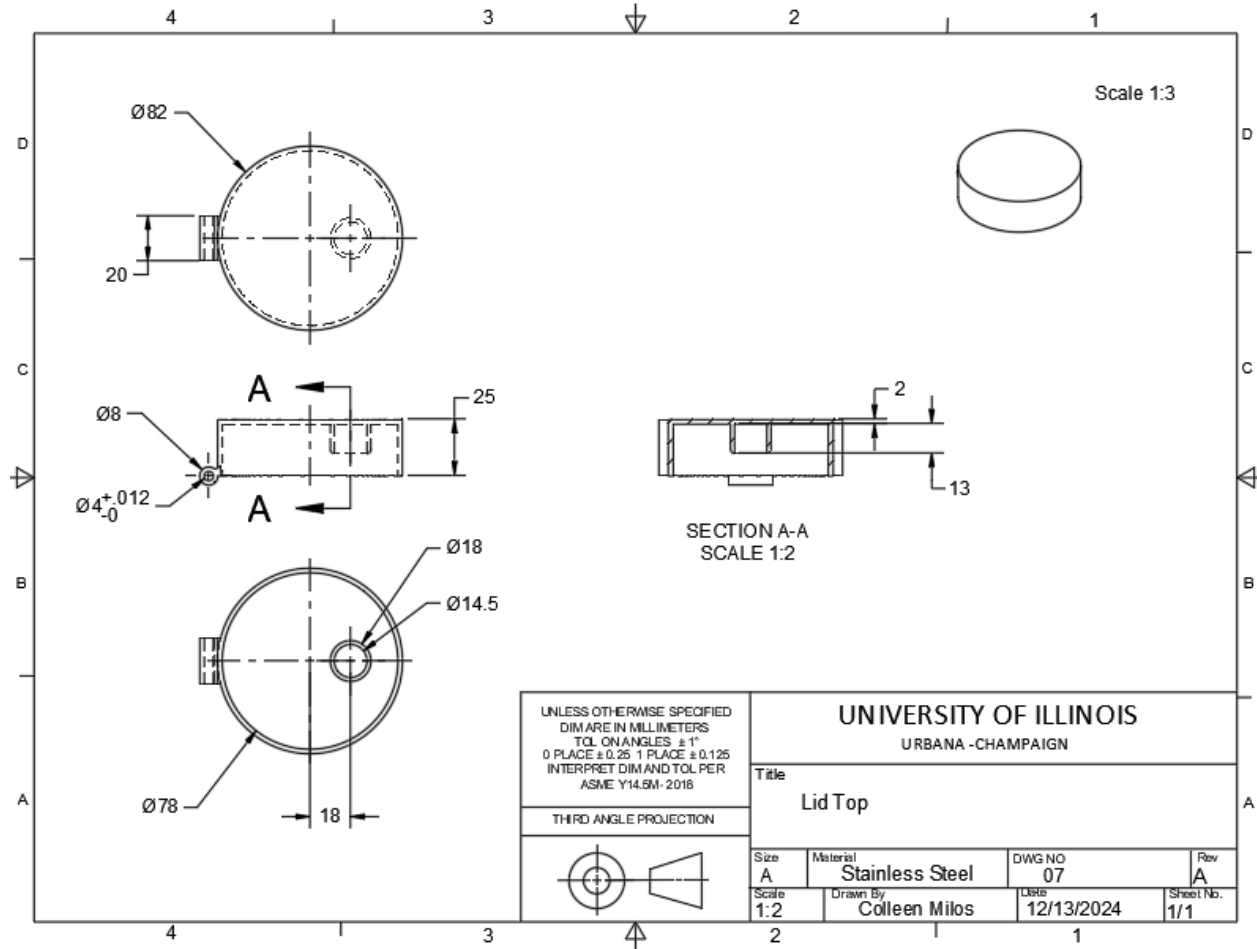
24

Figure 5:
Lid Bolt Drawing



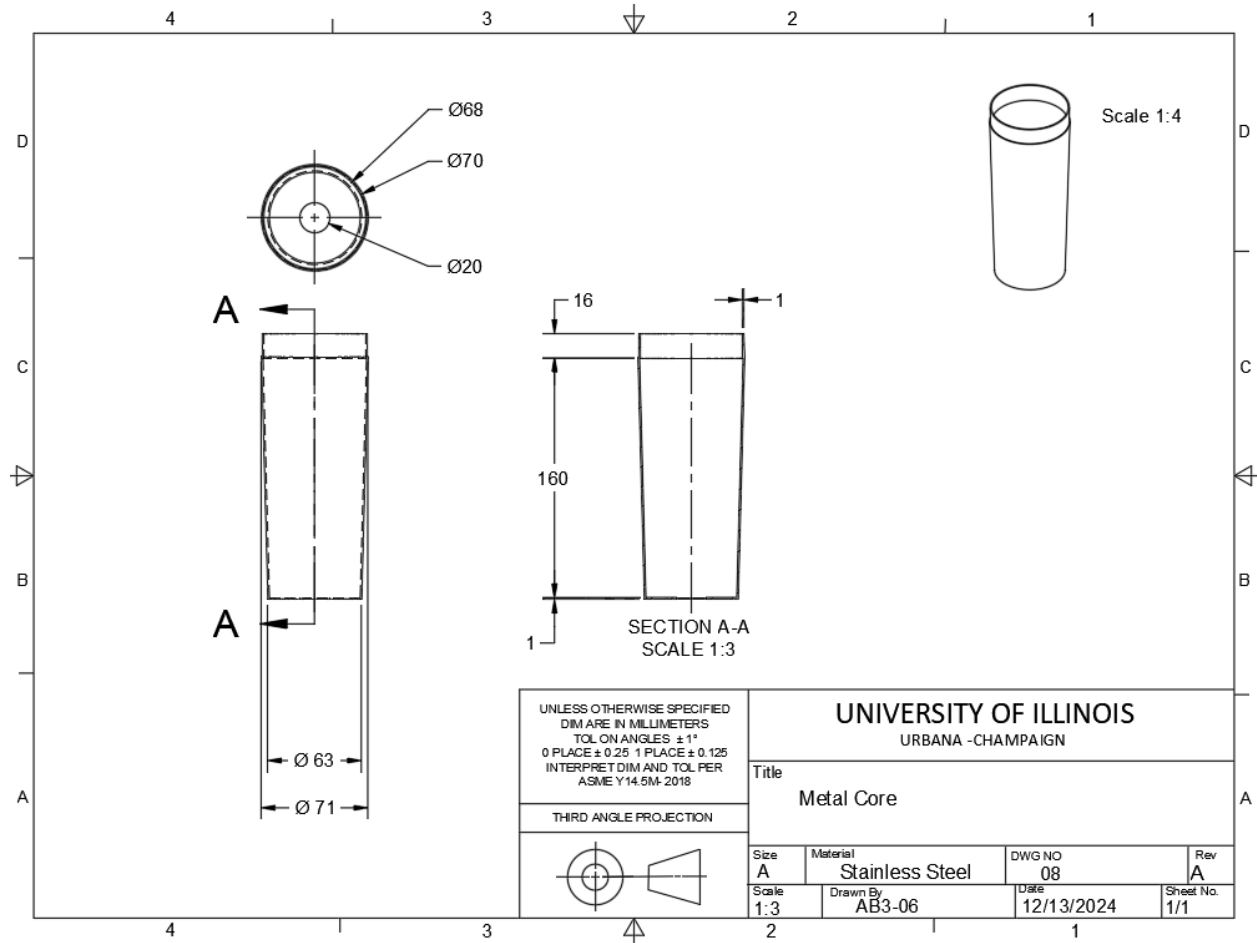
The lid bolt drawing is the part that allows for the rotating motion of the lid. It is attached to the bottom and top lid's hinge extensions, merging the two together. It is a simple design with a cylindrical core capped by two wider cylinders that stop it from sliding out.

Figure 6:
Lid Top Drawing



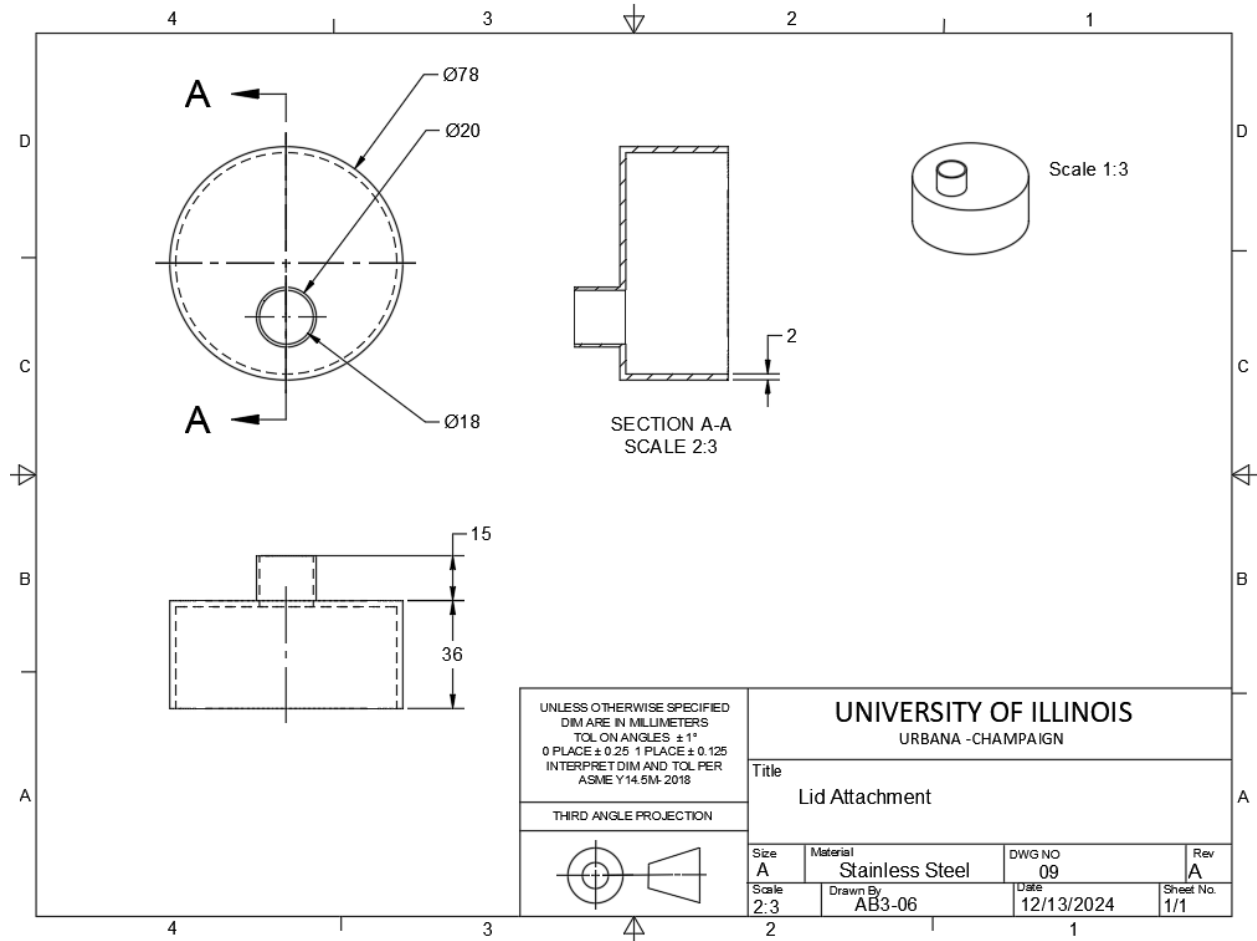
The lid top is the moving part in the assembly and it is attached by a small hinge extension to the bottom lid. The lid is dimensioned to allow for easy creation and assembly. The top lid features an extended hollowed out cylinder that, once closed, plugs the hole in the bottom lid. This feature ensures that no liquid is spilled out when the lid is closed and the cup is not being used.

Figure 7:
Metal Core Drawing



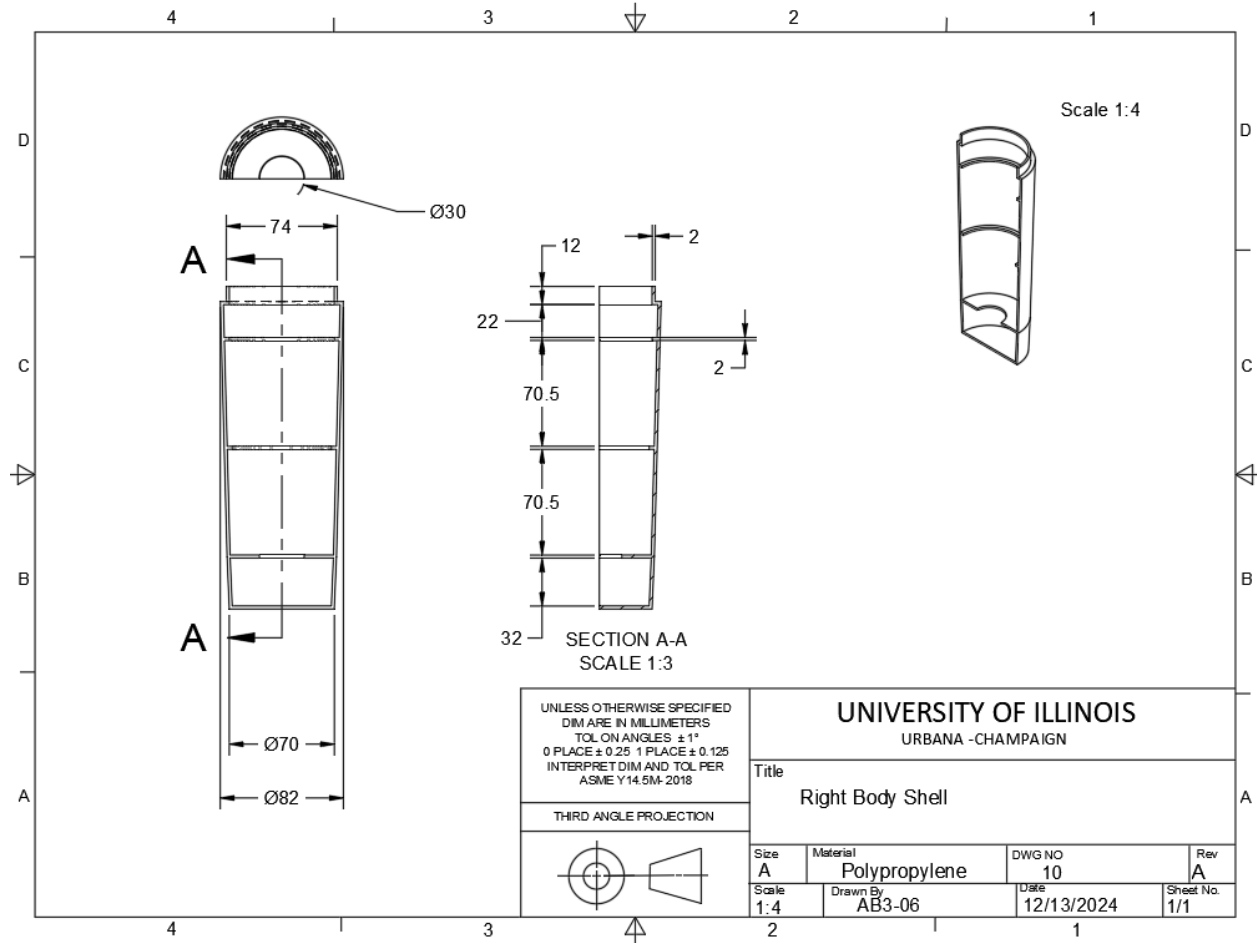
The metal core is the part that houses the actual liquid that the user would be drinking. It is dimensioned with the intent of fitting perfectly within the body shells' inner shelves. This separation from the outer edge of the body shell ensures that the likely hot metal core does not come in close contact with the user. The metal core features a small hole at its bottom that is left for the coil insert and casing to go through. Once the coil casing is inserted it is intended to plug this hole and stop all liquid from spilling out of the bottom.

Figure 8:
Lid Attachment Drawing



The lid attachment is the connection between the lid and the body shell. This part is attached to the inside of the bottom lid so that the liquid inside never makes contact with the plastic casing outside. This ensures compliance with food safety regulations and also helps the insulation of the drink within by adding more layers of material between the liquid and outside. This part also features a cylindrical exit that is fitted within the bottom lid's cylindrical exit and ensures the stability of the assembly and that the liquid avoids touching the plastic all the way until the end.

Figure 9:
Right Body Shell Drawing



The right body shell is essentially identical to the left body shell and is made by the same assembly methods and created for the same purpose. The reason it is a distinct part is for the purpose of the battery compartment at the bottom where the placement of electronic components would vary from shell to shell.

9.0 - TOLERANCE ANALYSIS

Within our design for the HotSip, there were few parts that required tolerancing according to standards through the ISO system. The lid of the HotSip rotates about a hinge joint, where both the top and bottom components of the lid are slid through a bolt, the Lid Bolt described in our assembly. All of these necessary tolerances are displayed below.

Figure 1:
Hole about Lid Bottom

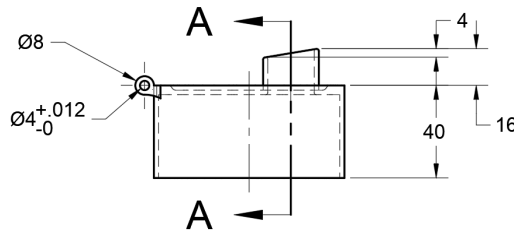


Figure 2:
Hole about Lid Top

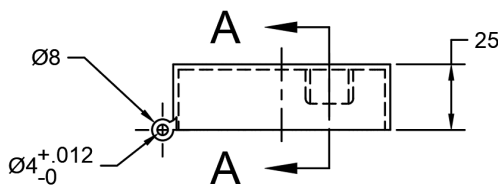
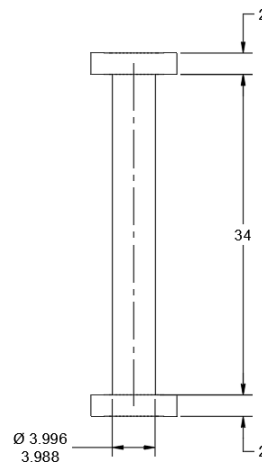


Figure 3:
Shaft within Lid Bolt



In Figure 1, we elected to utilize clearance fits when tolerancing the water bottle lid components and how they rotate about the bolt. Clearance fits, more particularly sliding fits, allow for accurate location and free rotation and movement about the hinge, and correspond to the ISO fit H7/g6. This will allow our water bottle lid to effectively open and close with ample range of motion.

To effectively calculate a tolerance analysis for the radial fit about the lid bolt and lid top, the IT grade proposed is a sliding fit (H7/g6), and the nominal dimension for both the lid bolt (shaft) and lid top (hole) is 4 mm. For H7, the tolerance lies between 0 to +0.012 mm. For g6, the tolerance lies between -0.012 and -0.004 mm. The clearance for a radial fit is equal to the largest hole - smallest shaft, or $4.012 - 3.988 = 0.024$ mm. The allowance about these mating parts for a radial fit is equal to the smallest hole - largest shaft, or $4 - 3.996 = 0.004$ mm. Because this part isn't equivalent to a bearing, we only need to calculate the inner diameter, as that's the only part that requires fitting and tolerancing.

An axial tolerance is not necessary for this part and for the part's calculation. The gap being analyzed would exist within the length 34 millimeters, the lid bolt's length between both ends, and the lengths of the lid top- and bottom-adhering mechanism to the hinge. As displayed below, there is no possible outcome for a production of these parts that would warrant axial tolerancing. While comparing figures 3, 4, and 5, it's apparent that there is a 0.5 mm gap between the lid top and lid bottom components,

and if they were mated with the Lid Bolt, there would be more than enough room disregarding tolerance. Additionally, the length of Lid Bolt not including the end extrusions is equal to 34 mm, which is well over the total length of 30.5, the Lid Bottom's mating component length.

Figure 4:

Lid Top view from top of part

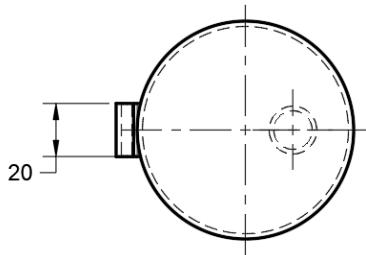
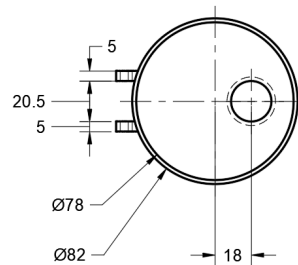


Figure 5:

Lid Bottom view from top of part



Because these measurements were taken and are so deeply extended as to not complicate axial tolerance measurements, there are none necessary for our part. We're anticipating free range of motion within the lid top, lid bottom, and lid bolt. This could change as the HotSip iterates through new revisions and improvements, but as of right now, no axial tolerancing is required to design or manufacture our part.

10.0 - MATERIALS, MANUFACTURING, AND COST ANALYSIS

10.1 - Bill of Materials

Figure 1:

Product's Complete Bill of Materials

Part #	Description	Material	Manufacturing Method	Fully Burdened Cost	Quantity	Total Part Costs	Investment Costs
1	Bolt for Lid	Steel, hot worked, AISI 1010	Stock Machining	\$0.65	1	\$0.65	\$388.96
2	Bottom of Lid	Polypropylene, Med-Impact Copol	Plastic Molding	\$0.43	1	\$0.43	\$14,747.16
3	Coil Casing	Stainless Steel, Stock 15-5PH	Sheet Metal: Progressive Die stamping	\$1.37	1	\$1.37	\$26,102.57
4	Coil Insert	Stainless Steel, Stock 15-5PH	Sheet Metal: Progressive Die stamping	\$0.35	1	\$0.35	\$27,589.20
5	Coil	Copper (ID: 8873K11) McMaster-Carr	Purchased	\$2.94	1	\$2.94	\$0.00
6	Metal Core	Stainless Steel, Stock 15-5PH	Sheet Metal: Progressive Die stamping	\$2.03	1	\$2.03	\$42,545.07
8	Body Shell	Polypropylene, Extrusion Sheet	Injection Molding: Blow Molding	\$0.41	2	\$0.82	\$14,660.51
9	Top of Lid	Polypropylene, Med-Impact Copol	Plastic Molding	\$1.88	1	\$1.88	\$13,832.51
10	Lid Attachment	Stainless Steel, Stock 15-5PH	Sheet Metal: Progressive Die stamping	\$2.59	1	\$2.59	\$32,975.77
TOTALS						\$13.06	\$172,841.75

(Note: volume discounts assumed)

10.2 - Materials and Manufacturing Analysis

When designing our parts for manufacture we considered numerous different details including ideal materials, manufacturing methods, and our product's intended selling price.

When selecting the ideal materials for our products' various parts, it was crucial that we considered constraints placed on us due to the nature of our product; the product will, for instance, contain components that will be heated to high temperatures, contain electrical components, and also hold a liquid whose purpose is human consumption. We thus had to consider the safety and ability of our chosen materials to comply with these respective areas.

Our products metal components were designed with Stainless Steel (Stock 15-5PH); this form of stainless steel was chosen for our product due to its lack of corrosivity and ability to withstand high temperatures; it was also one of the cheapest forms of stainless steel, which made it incredibly effective at lowering material costs. These components were manufactured with a Sheet Metal Progressive Die Stamping process; while this manufacturing process would have high tooling costs (totalling close to

\$130,000), this is a worthy investment as we are designing with the intention of producing roughly one million parts per year. With such a high volume of production, it makes sense to invest in a hard-tooling process like progressive die stamping that results in each part being extremely cheap (greatly lowering the manufacturing cost per part).

Our products' plastic components were designed with Polypropylene (Med-Impact Copol); this material was selected due to its use in various other similar products (used by similar companies like Stanley Cup) and its ability to withstand heat. As most parts of the product the user directly interacts with are made from this material, it was also important that we selected a material the user would be comfortable touching and handling. These components were primarily designed with either a Plastic Molding or Injection Molding manufacturing process; both of these processes are forms of hard tooling that, while raising initial tooling costs greatly, lower the cost per manufactured part. Such an investment is necessary as we are designing to produce roughly one million parts per year,

By selecting materials that comply with the safety and comfortability constraints presented by our product, as well as selecting manufacturing processes that lower the costs to produce each unit, we successfully attained a safe product that is significantly lower than our intended selling price of roughly \$60-75.

11.0 - CONCLUSION

We believe that our product could go to market, however, there are still many steps we would need to take before pitching our product to investors or getting our product on retailers' shelves. Despite this, we believe that when/if our product does go to market we predict that there would be medium risk associated with our product. Our product would have to compete with already very popular water bottle/thermos brands like Owala, Stanley, and Yeti. However, our product, HotSips, the main difference is its heating capabilities, which is nonexistent in our competitors product line ups.

Our total product tooling investment is \$172,841.75 and our product's components minus the electrical components (battery) is \$11.92. Due to this we believe that our product can be priced at around \$60-\$75, which remains below our competitors' (other heated water bottle brands) price range. However, it is more expensive than typical popular water bottle brands, but other popular water bottles don't have the same heating capacity that our product would. Our total profit, with taking the battery into consideration, could be close to \$40 for every unit sold.

For future steps, we should consider engaging in conversation with other water bottle distributors to gain insight on their recommendations and their considerations that they underwent in their design process in order to gain a better understanding of what made their water bottle popular and stand out amongst a competitive market. Further, for our product we would need to consult with other engineers, specifically electrical engineers, to help us develop and manage the battery / electrical components of our product. The largest knowledge gap that we currently have with our product remains on the electrical components required to make our product work. That is why it may be prudent to consult with various electrical engineers to help fully flesh out our product. In terms of funding for the development of our product, we could find investors that like the idea of our product and are willing to help us fully flesh out the idea. Another way is to sell the idea to an already established and profitable company to help fund our goal for HotSip as a new product in their line of products. In terms of prototyping, we could 3D print the different components of our product and make sure that the assembly of our product works.

As we wrapped up our project, we began discussing avenues of improvements and future iterations for our product HotSip. One of the main things we discussed was taking a closer look at what makes popular water bottle brands so appealing to consumers and how we can make the HotSip the next 'microtrend' for consumers. We believe that that element is heavily centered around the design of the product. Further, our product can also have various different iterations based on size/amount of liquid that our product can hold. In addition, we have considered adding another attachment that we had ideated during ideation; an sleeve for our drink that has a handle. In conclusion, our product is very versatile and can have multiple different styles and sizing that would make it very marketable and desirable. While there is still medium risk to our product to go to market, we believe that with the correct marketing and the difference between our product and others could help us have a profitable product.